Project Report: “Quote Attribution To Source Book Challenge”

Lior Malachi 322409376 Eliezer Mashihov 207476763

Project Overview & Task Definition

This project focuses on a deceptively simple but challenging task: **Given a literary quote, identify the book from which the quote was taken.** This task requires models to rely entirely on their internal knowledge and memory, without access to external tools, databases, or search engines. It therefore evaluates LLM's intrinsic understanding of literature - its ability to connect a quote’s tone, phrasing, and thematic cues to its correct source - rather than its capacity to retrieve information from the web. (our research questions – on **Appendix 1**)

Dataset Creation & Characteristics

Implemented in generate_data.py, our **automated pipeline** builds the dataset directly from Goodreads, ensuring transparency, scalability, and reproducibility. It consists of three main stages: (scraping is parameterized for versatility, more in **Appendix 2**)

Characteristics: (see **Appendix 3** for full dataset statistics)

- 16,702 quotes of various difficulty levels, taken from 289 books written by 211 authors.
- **Rich metadata** per example (difficulty score, rarity, entities, etc.), see **Appendix 4**.
- Diverse genres, linguistic style and eras.

This design ensures both **breadth and interpretability**: models face not just random samples, but systematically varied challenges.

Stage 1: Data Collection

- Scrapes 300 books from Goodreads’ *“Best Books Ever”* list.
- For each book, extracts title, author, and metadata - ratings, reviews, publication year, page count.
- Gathers up to two pages of quotes per book, producing thousands of literary examples across genres and eras.

Stage 2: Features Calculation, Metadata Generation and Difficulty Scoring

For each quote, a set of calculated linguistic and contextual features is derived to capture the **quote’s complexity and uniqueness** (also serve as **metadata**). Each feature captures an aspect of the quote’s distinctiveness or recognizability and contributes numerically to its final score. This score will determine the difficulty level of the sample. **Positive values → harder**, **negative values → easier**. See explained features table at **Appendix 1**.

Stage 3: Classification to Difficulty Levels

The total score is computed by summing the contributions of all features.
Quotes are then divided into 3 difficulty levels: **Easy:** score ≤ –3, **Medium:** –3 < score ≤ 0, **Hard:** score > 0. (More about that on **Appendix 5**).

Evaluation Methodology

Implemented in evaluation.py and run_eval.py. We designed an evaluation that goes beyond a simple accuracy score by analyzing to why a model fails, correlating its performance with our rich metadata.

(a) Balanced sampling: Loads the dataset (data/examples.json) and draws a fixed number of samples **(we used 150)** from each difficulty level - easy, medium, and hard. This ensures that all difficulty types are **equally represented in the evaluation**.
(b) Multi-Model Benchmarking: Our code supports testing multiple models through the litellm API, including: OpenAI GPT-5 series, Google Gemini 2.5 family, Anthropic Claude Sonnet, and more. Each model receives identical prompts.

(c) Judge-LLM Evaluation: We used LLM-as-a-Judge for evaluation. The reliable Grok 4 Fast, acting as an evaluator(binary **Yes/No** decisions).

Results

(see additional tables and full results at **Appendix 6,** including explanations)

Best performing mode: **gemini 2.5 pro** with **56.44% overall accuracy** on the dataset. Worst performing mode: **gpt 5 mini**: 29% overall accuracy.

Overall Model Performance (Selected Model Performance Across Difficulty Levels)

Accuracy	Overall (%)	Easy (%)	Medium (%)	Hard (%)
gemini 2.5 pro	56.44	77.33	62.67	29.33
gpt 5 mini	29.56	43.33	37.33	8
claude sonnet 4	48.44	70.67	56.67	18

Feature Correlation (Correlation of Metadata Features with Model Accuracy)

avg accuracy per difficulty level	Easy	Medium	Hard	Correlation
accuracy (%)	62.2	49.8	19	**-1.0 ($p < 0.001$)**

	difficulty score	book popularity	entities count	quote length	lexical rarity	publication year
correlation	-0.40	0.26	0.22	0.22	0.03	-0.24
p value	**$p < 0.001$**	**$p < 0.001$**	**$p < 0.001$**	**$p < 0.001$**	$p < 0.05$	**$p < 0.001$**

Analysis

run_eval.py script automates all post-evaluation analysis by computing a suite of quantitative metrics, including granular **accuracy breakdowns by metadata**, **model response times**, and statistical analyses like **correlations** and **significance** testing. (see charts in appendices)

LLMs perform well on quotes from popular, culturally significant books, but as the quotes become more obscure or linguistically complex, we see a clear pattern of "near miss" errors. These include misattributions to other works by the same author or books from a similar genre, such that the hardest cases expose the limits of the models' static, internal knowledge base. The detailed results for each quote are saved, allowing for a granular analysis of these specific failure modes. (interesting observations in **Appendix 7**).

Limitations

Our study's primary limitations stem from the dataset's scope and source. By scraping from the top pages of a single, highly popular Goodreads list ("Best Books Ever"), our data is inherently biased towards well-known, predominantly English-language fiction. This popularity bias is further amplified by our method of extracting the most-liked quotes for each book. Consequently, our findings may not generalize to other genres like non-fiction or poetry, where different linguistic patterns and the challenge of intertextuality -one work quoting another, could be more prevalent. (**Appendix 12** for interesting unanswered questions left)

Conclusion & Future Work Suggestions

We developed a robust framework to evaluate LLM performance on a closed-book quote attribution task, using a metadata-driven difficulty score. Our results show that while models excel at identifying popular quotes, they struggle with obscure or linguistically complex passages, revealing a heavy reliance on memorized training data over deep semantic inference. Future work could explore the models' reasoning using Chain-of-Thought prompting, test the findings' generalizability with datasets from new genres and languages, and directly address the "closed-book" limitation by evaluating models within a Retrieval-Augmented Generation (RAG) framework to improve performance. (see **Appendix 11**)

Appendices

1. Our research questions list:

- How accurate are LLMs at closed-book quote attribution?
- Which quote characteristics (e.g., length, popularity, named entities) most significantly impact model performance?
- Does our engineered difficulty score strongly correlate with model accuracy?

1.1 Challenges our dataset presents:

Our quote attribution task provides a unique challenge by serving as a "closed-book" exam that:

- Checks the LLM's specific factual knowledge, not just its general reasoning.
- Success requires the model to navigate the complex interplay between direct recall (relying on passages memorized from its training data) and semantic inference (using contextual clues like genre or style to deduce the source).
- This is further complicated by the "signal vs. noise" problem, where the model must identify the unique stylistic fingerprint of a book while ignoring generic thematic elements that could misleadingly point to other works in the same genre.

2. Scraping Parameterization Details

We detail below the specific parameters used in this project to generate the raw dataset from Goodreads:

- **BASE_LIST_URL:** The source URL on Goodreads for the "Best Books Ever" list, which served as the entry point for discovering potential books.
- **MAX_LIST_PAGES**: The number of pages crawled on the source list. This determined the size of the initial pool from which books were randomly selected.
- **MAX_BOOKS_TO_PROCESS:** The total number of unique books randomly selected from the discovered pool for full processing (metadata and quote scraping).
- **MAX_QUOTE_PAGES:** For each of the 300 books, the maximum number of quote pages scraped. This parameter controlled the depth, or number of samples, per book.
- **POLITENESS_DELAY_S:** A 3-second delay between each HTTP request to the Goodreads server. This was implemented to ensure responsible scraping and avoid rate-limiting.

3. Dataset statistics

Characteristic	easy	medium	hard	Overall
Count_of_Quotes	1873	7998	6831	16702
Avg_Difficulty_Score	-3.71	-1.66	1.84	-0.46
Avg_Quote_Length	70.56	55.31	17.06	41.38
Avg_Quote_Likes	1649.45	528.35	394.31	599.25
Avg_Rarity_Ratio	0.11	0.07	0.05	0.06
Avg_Entities_Count	8.35	4.7	0.3	3.31

4. Metadata Fields Description

These fields were used to calculate the difficulty score and to drive the analysis of model performance.

- **difficulty:** The final difficulty classification of the quote: "Easy", "Medium", or "Hard".
- **score:** The raw numerical difficulty score. Negative values indicate an easier quote, and positive values indicate a harder one.

Quote metadata:

- **length_words:** The total number of words in the quote text.
- **sentences:** The number of sentences in the quote text.
- **entities_count:** The number of potential named entities (e.g., character names, places) found within the quote.
- **rarity_ratio:** A ratio indicating the proportion of lexically rare (infrequent) words in the quote.
- **stopword_ratio:** A ratio indicating the proportion of common stopwords (e.g., "the", "a", "is") in the quote.

Popularity metadata:

- **likes:** The number of "likes" the specific quote has received on Goodreads.
- **ratings_count:** The total number of ratings the source book has received on Goodreads, indicating its overall popularity.
- **reviews_count:** The total number of text reviews the source book has received on Goodreads.

Source book metadata:

- **publication_year:** The year the source book was first published, indicating its literary era.
- **page_count:** The number of pages in a common edition of the source book.

5. Dataset sample difficulty

The purpose of this measure is to assign a single numerical score to each quote that quantifies how difficult it should be for a language model to identify. It does this by analyzing several key characteristics, or "features," of the quote and its source.

5.1 Difficulty Features Table

Feature	**Description & Interpretation**	**Effect on Score**
Quote Length	Short quotes provide little stylistic or contextual information, making attribution harder.	+1.5 (<=10), +0.5 (11-25), –1.0 (>25)
Likes (Quote Popularity)	Unpopular quotes are rarely seen in training data, making them harder to recall. Highly popular quotes are likely familiar to LLMs.	+1.5 (<1k), +0.5 (1k-10k), –1.0 (>10k)
Named Entities Count	Named entities (e.g., character or place names) provide strong contextual anchors. Quotes with none are harder to link; those with several are much easier.	+0.5 (0 entities), –1.5 (1–2 entities), –2.5 (>2 entities)

Sentences Count	Multi-sentence quotes supply additional stylistic and narrative cues. Single-sentence quotes are brief and ambiguous, increasing difficulty.	0.0 (1 sentence), –0.25 (2 sentences), –0.5 (3+)
Stopword Ratio	High ratios of stopwords indicate generic language, offering few unique tokens for recognition. Lower ratios mean more distinctive wording.	+0.5 (high >0.62), –0.5 (low <0.45)
Lexical Rarity	Rare content words make a quote more identifiable because they capture author-specific phrasing. **Generic vocabulary reduces uniqueness**.	–1.0 (high rarity), –0.5 (medium), +0.5 (low rarity)
Book Popularity (Normalized)	Highly popular books make quotes easier. book_pop_norm is taking log1p-scaled average likes of top 10 quotes and normalize across all books using min-max scaling ([0,1] value)	Adjustment = 3 × (0.5 – book_pop_norm)

5.2 Explaining the difficulty score function

The function **compute_score** is responsible for calculating a quote's difficulty score. It quantifies a quote's difficulty by treating it as a balance between its popularity and its distinctiveness. Features related to fame, such as a high number of "likes" on Goodreads or belonging to a very popular book, heavily reduce the score, making the quote easier on the assumption that the model has likely "memorized" it from its training data.

The quote's intrinsic "fingerprint" is also critical. Longer quotes, those with multiple sentences, and especially those containing unique named entities (like character or place names) are also scored as easier, as they provide more context and unique identifiers for the model to latch onto. Conversely, short, generic quotes from unpopular books are penalized with a higher (harder) score.

5.3 Difficulty Category Rationale

A positive final score signifies that features contributing to obscurity and genericness have outweighed the factors of fame and distinctiveness, thus classifying the quote as "Hard."

The difficulty thresholds were chosen to create distinct, meaningful categories. The 0 threshold acts as a neutral baseline; a score above zero means features contributing to difficulty (like obscurity) outweigh features contributing to simplicity (like named entities). The -3 threshold was set empirically to isolate unambiguously easy quotes, as a score this low requires a quote to have a combination of multiple strong "easy" features, ensuring it's not just marginally simple.

6. Full results

6.1 Insights

- Gemini models show consistently better results on all difficulty categories than the rest of the models. We can assume the Google is an expert in information retrieval, and hence in its LLM training process knew how to leverage that realm into their models.
- Average accuracy per difficulty level was calculated by averaging the accuracy measure on the responses of all models on each difficulty category samples.
- All of our meta-data features are ordinal. So in this way, we can truly analyze a trend as the feature value / category "increases" (for example lexical rarity, named entities, and other features).

- We used Pearsons test to calculate correlation between the categories' ordinal but unevenly spaced variables.
- All models have shown a decreasing trend in accuracy as difficulty increases.
- We found perfect and significant negative correlation across models on difficulty categories.
- The accuracy for easy samples ranges from 43% to 77%, for medium samples ranges from 37% to 62%. And for hard samples ranges from 8% to 29% across the models.
-

6.2 Overall model performances:

Model	Overall Accuracy (%)	Easy (%)	Medium (%)	Hard (%)	Avg Response Time (s)
gemini 2.5 pro	56.44	77.33	62.67	29.33	11.9
gemini 2.5 flash	45.33	63.33	51.33	21.33	0.65
gemini 2.5 flash lite	32.67	49.33	38.67	10	0.69
gemini 2.0 flash	44.89	61.33	52	21.33	0.8
gpt 5 mini	29.56	43.33	37.33	8	16.53
claude sonnet 4	48.44	70.67	56.67	18	4
grok 4 fast	43.33	64	50	16	17.66
deepseek r1	48.89	68.67	50	28	19.15

6.3 Metadata Feature Performance Gap Table

Each model-feature cell value has the largest change in accuracy across its categories.

Feature (Impact)	claude sonnet 4	deepseek r1	gemini 2.0 flash	gemini 2.5 flash	gemini 2.5 flash lite	gemini 2.5 pro	gpt 5 mini	grok 4 fast
Lexical Rarity	8.2	-5.5	1.6	9.9	0.5	6.6	16.5	11.5
Publication Year	-45.2	-41.6	-40.4	-44.8	-41.3	-35.6	-32.5	-31.5
Named Entities	45.8	28.2	33.9	35.5	38.1	38.6	35	43
Book Popularity	26.7	22.7	24.7	24.7	23.2	31.5	21.1	22
Quote Length	63.8	36.5	43.7	42.1	47.4	54.7	38.8	61.5
Difficulty Score	59.1	37.9	49.8	47.9	45.5	58.1	39.5	48.6

7. More observations

- We observed common failure modes where multiple models converged on the same incorrect answer by anchoring on a single, misleading keyword from the quote.
- Many incorrect responses were "near misses" that shared semantic themes or keywords with the correct title, indicating partial knowledge but a failure in precise attribution.

- At its core, this task tests a model's fundamental ability to form a semantic association between a quote's content and its source's title, moving beyond simple recall.
- The low accuracy across all models, even on 'Easy' samples, demonstrates that closed-book quote attribution is an inherently challenging task for current LLMs.
- The difficulty classifications are based on a configurable heuristic, and their thresholds could be adjusted in future work to probe different performance aspects.
- Our evaluation's primary strength is its multi-faceted analysis across six core metadata features, providing specific insights beyond the three abstract difficulty levels.

8. Dataset Sample Examples


```json
{
    "id": "quote_1",
    "prompt": "From which book is the following quote?\n\n\"“You never really understand a person until you conside
    "quote": "“You never really understand a person until you consider things from his point of view... Until you c
    "answer": "To Kill a Mockingbird",
    "author": "Harper Lee",
    "metadata": {
        "difficulty": "easy",
        "score": -5.323,
        "likes": 18357,
        "length_words": 27,
        "entities_count": 1,
        "page_count": 323,
        "publication_year": 1960,
        "ratings_count": 6745083,
        "reviews_count": 129678,
        "length_bin": ">=26",
        "likes_bin": ">=10k",
        "entities_bin": "1-2",
        "sentences": 3,
        "stopword_ratio": 0.423,
        "stopword_bin": "<0.45",
        "rarity_ratio": 0.0,
        "rarity_bin": "<0.15",
        "book_pop_norm": 0.941,
        "book_pop_adj": -1.323
    }
```

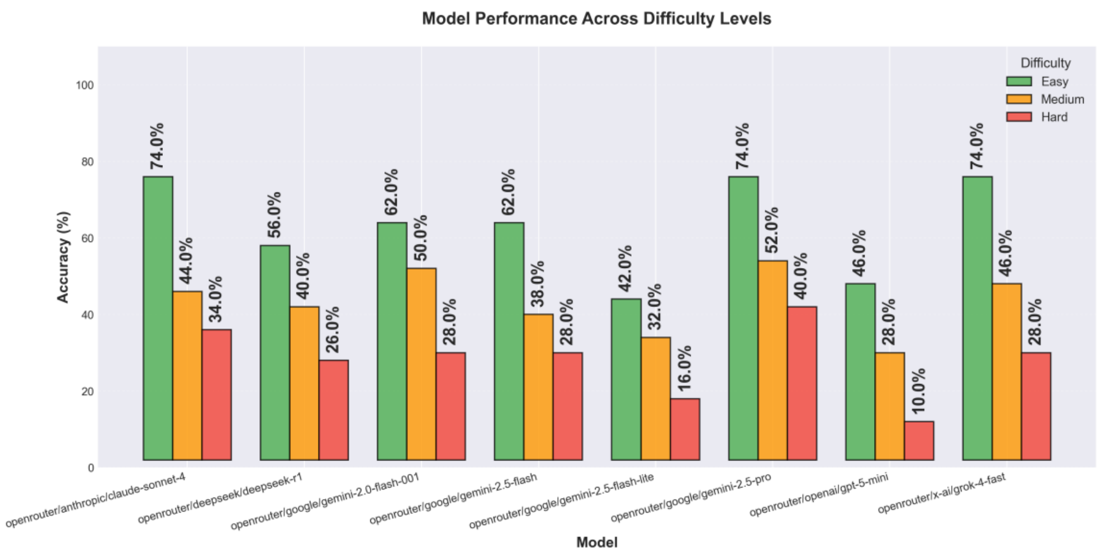

9. Detailed Metadata Performance Breakdown

Feature	Category	count	claude sonnet 4	deepseek r1	gemini 2.0	gemini 2.5	gemini 2.5	gemini 2.5	gpt 5 mini	grok 4 fast
Publication Year	Pre-1950	114	71.1	71.9	64.9	69.3	53.5	73.7	47.4	60.5
Publication Year	1950-1980	78	60.3	57.7	52.6	55.1	44.9	66.7	41	48.7
Publication Year	1980-2000	101	48.5	44.6	47.5	41.6	31.7	57.4	23.8	41.6
Publication Year	Post-2000	155	25.8	30.3	24.5	24.5	12.3	38.1	14.8	29
Lexical Rarity	Low Rarity (Generic)	364	47.5	48.1	44.5	45.6	32.7	56.6	30.8	43.7
Lexical Rarity	Medium Rarity	58	58.6	51.7	48.3	48.3	32.8	58.6	29.3	46.6
Lexical Rarity	High Rarity (Unique)	28	39.3	53.6	42.9	35.7	32.1	50	14.3	32.1
Named Entities	No Entities	120	21.7	32.5	25.8	24.2	10	34.2	9.2	19.2
Named Entities	Few Entities (1-2)	124	42.7	45.2	38.7	41.9	29	50.8	25	35.5
Named Entities	Many Entities (3+)	206	67.5	60.7	59.7	59.7	48.1	72.8	44.2	62.1
Book Popularity	Bottom Third (Unpopular)	151	42.4	42.4	39.1	38.4	29.1	47.7	23.8	37.7
Book Popularity	Middle Third (Moderate)	150	34	39.3	32	34.7	16.7	42.7	20	32.7
Book Popularity	Top Third (Popular)	149	69.1	65.1	63.8	63.1	52.3	79.2	45	59.7
Quote Length	very_short	37	13.5	32.4	24.3	24.3	8.1	24.3	10.8	10.8
Quote Length	short	91	26.4	28.6	26.4	25.3	14.3	36.3	11	22
Quote Length	medium	118	39	41.5	33.9	38.1	28.8	47.5	26.3	31.4
Quote Length	long	85	60	60	56.5	56.5	36.5	72.9	34.1	56.5
Quote Length	very_long	119	77.3	68.9	68.1	66.4	55.5	79	49.6	72.3
Difficulty Score	Easy (Score < -3)	150	70.7	68.7	61.3	63.3	49.3	77.3	43.3	64
Difficulty Score	Medium (Score -3 to 0)	150	56.7	50	52	51.3	38.7	62.7	37.3	50
Difficulty Score	Hard (Score 0 to 3)	124	19.4	27.4	23.4	22.6	11.3	31.5	8.9	16.1
Difficulty Score	Very Hard (Score > 3)	26	11.5	30.8	11.5	15.4	3.8	19.2	3.8	15.4

We can see most metadata features have a significant count of samples, that the accuracy of the models is calculated on. Except the category of very hard difficulty score, that has the lowest sample count, even though it is 26.

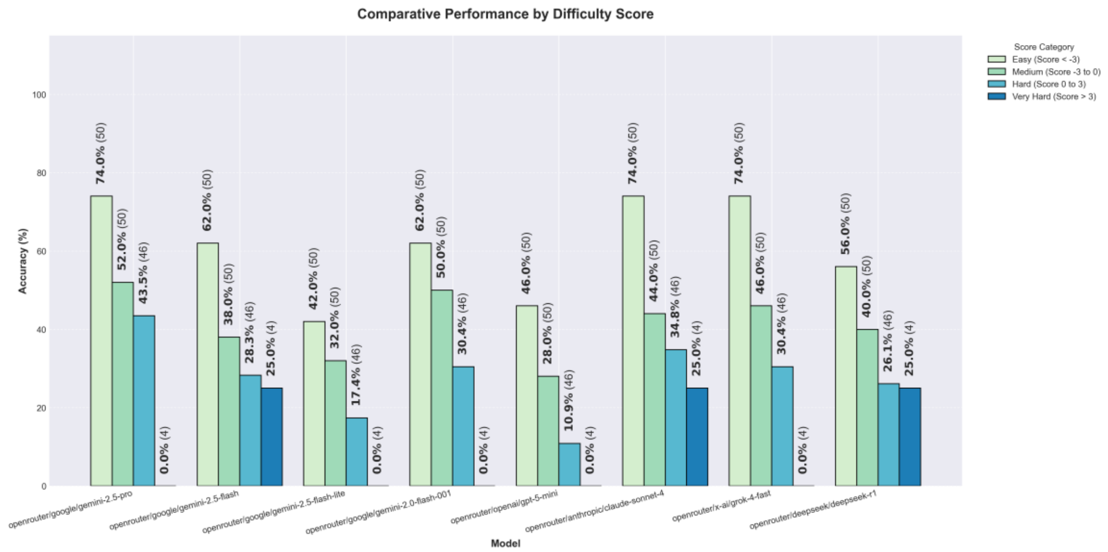
10. More charts for detailed drill-through analysis

10.1 Model performance across difficulty levels

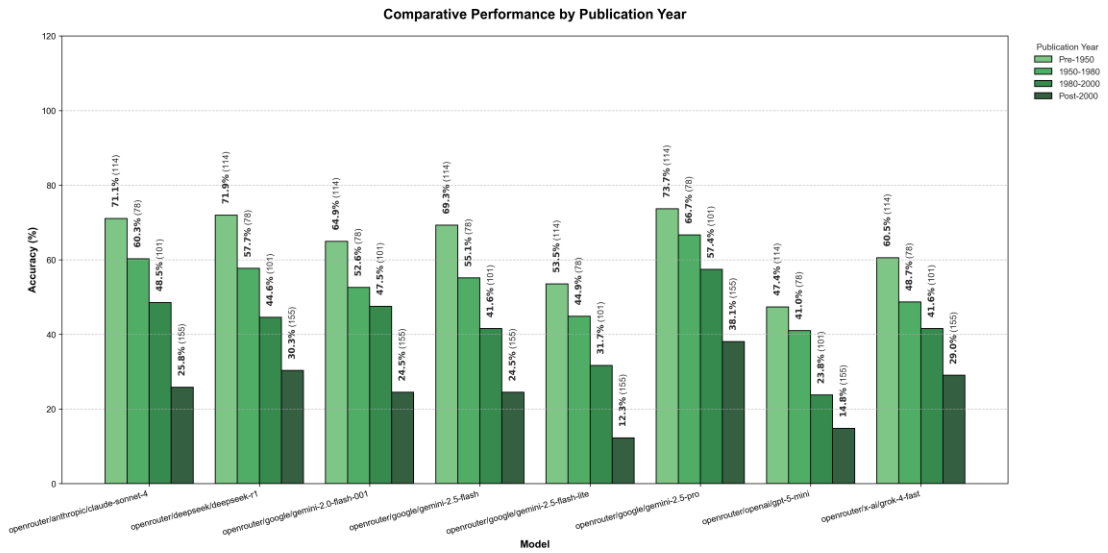


10.2 Accuracy by difficulty score

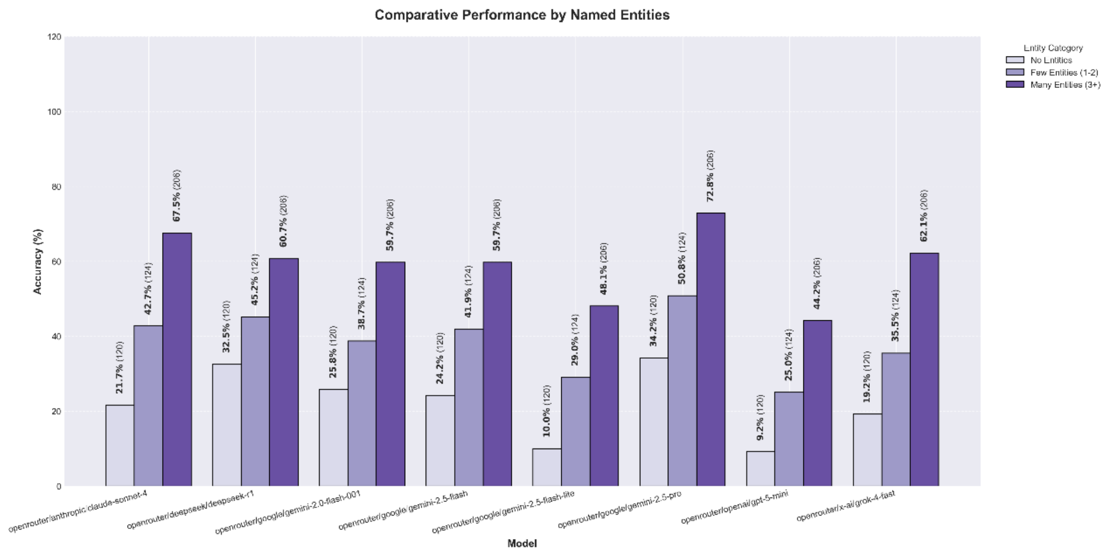
In our analysis, we introduced a fourth category for difficulty which is “extremely hard”, for samples that have a difficulty score of 3+. Those samples most have many features that together make them very difficult to solve. Checking the accuracy for those samples is letting us check the case where many factors contribute to a high difficulty score, and if the accuracy drops, it means our measure formula is reliable. Charts are attached for analysis. We can see clear trends in almost every meta-data feature we created.



10.3 Accuracy by publication year



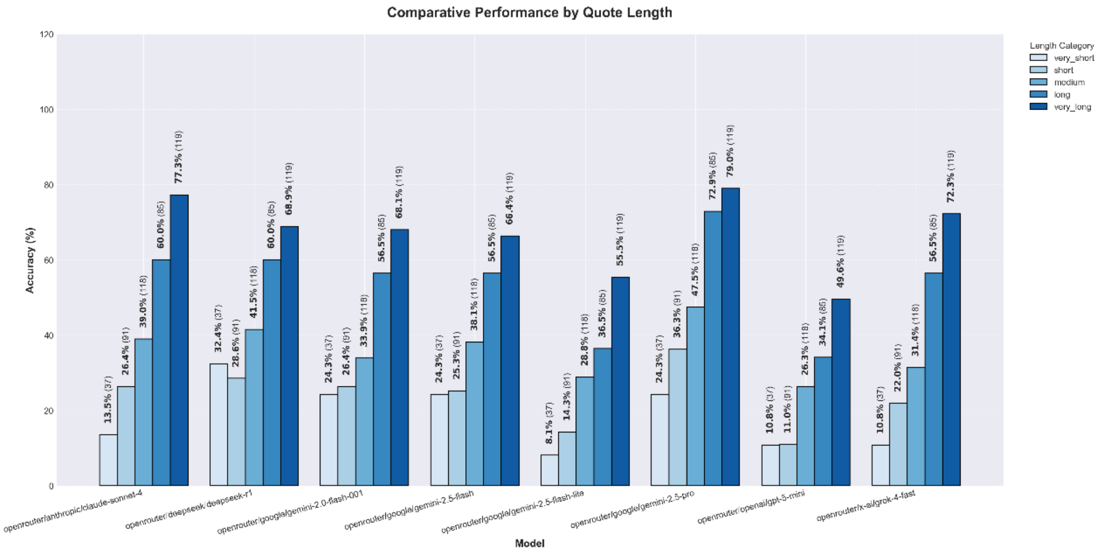
10.4 Accuracy by named entities



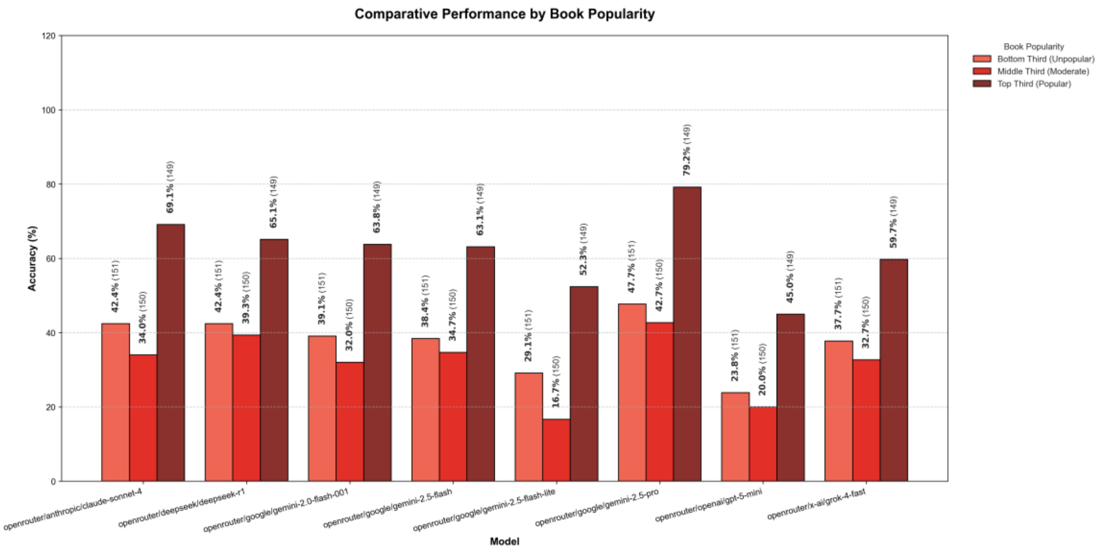
10.5 Accuracy by length of quote

We can see a nice anomaly in the trend here, where “very short” quotes have higher accuracy than “short” quotes, and the increasing trend is consistently beginning from the “short” to the “very long” length of quote.

very_short: < 50 characters (brief snippets) short: 50-100 characters (sentence fragments) medium: 100-200 characters (full sentences) long: 200-300 characters (multiple sentences) very_long: > 300 characters (paragraphs).

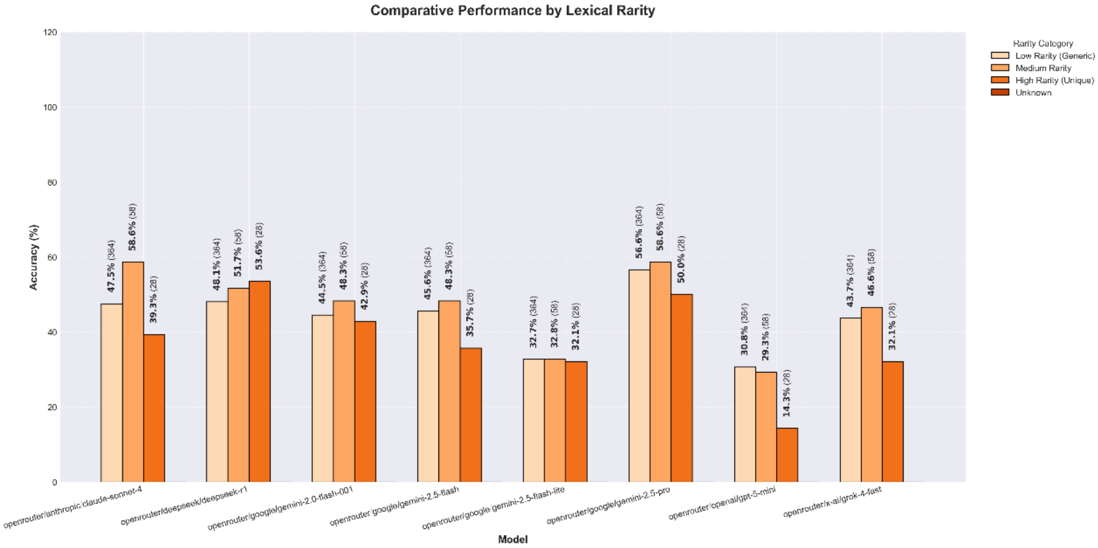


10.6 Accuracy by book popularity



10.7 Accuracy by lexical rarity

Our initial hypothesis, which was built into the difficulty scoring rubric, predicted that a higher proportion of rare words would make a quote easier to identify. The reasoning was that rare words would act like unique keywords, creating a distinct "fingerprint" that a model could more easily associate with a specific source, much like a good search engine query. However, our empirical results revealed the opposite. The correlation analysis showed a weak but statistically significant negative correlation (r = -0.08, p < 0.001), meaning that as the lexical rarity of a quote increased, the models' accuracy tended to decrease.



11. Additional future research ideas & experiment improvement

We find our project very interesting and it has quite a few open leads that can be further explored. Few that we have thought of are:

Advanced prompt engineering:

- Prompt Phrasing: Systematically investigate how variations in the prompt's phrasing impact model accuracy.
- Contextual Hinting: Test whether providing a hint in the prompt, such as the book's genre, measurably improves performance.
- Few-Shot Learning: Evaluate if model accuracy improves with k-shot prompting, where the prompt includes a few examples of correct quote-to-book attributions.

Cross lingual and Cross genre:

- New genres: Replicate the experiment on datasets from different genres (e.g, non-fiction, poetry, scientific papers) to see if the observed performance patterns hold true across different writing styles.
- New languages: Build similar datasets in languages other than English to explore how a language's representation in the training data correlates with model performance on this task.

Cross media adaptation: Adapt the methodology to other domains, such as attributing famous quotes to popular movies or historical speeches to their speakers.

Targeted dataset curation: Refine the scraping process to create more targeted datasets for specific analyses, such as gathering 100 "Very Hard" quotes or 100 quotes that contain no named entities, to allow for more focused stress-testing of the models.